



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 9 • STANDARD 6.AT.11

Analyze Graphs of Relationships Between Two Variables

Lesson 9-2 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period

My Notes

Level 2 Standard



TODAY'S OBJECTIVES

Content Objective: I can use a table and a graph to determine and analyze the relationship between two variable quantities.

Language Objective: I can describe what a point on a graph represents using the words ordered pair, axis, and coordinates.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

graph

table of values

ordered pair

axis

coordinates

Rate of change

Origin



Tap any word to see its meaning and a picture.

1

A picture on the coordinate plane showing how two quantities relate is a

2

A list of matched pairs that shows a relationship in numbers is a

3

Two numbers written (x, y) that name one point are an

4

Each number line framing a graph, labeled with a quantity, is an

5 The numbers in an ordered pair that tell how far to move are its

.

6 How much y changes when x goes up by one is the .

7 The point $(0, 0)$ where the axes meet is the .

Write About the Math

What observations can you make about the data using the graph?

¿Qué observaciones puedes hacer sobre los datos usando la gráfica?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **graph**
gráfica

☐ **table of values**
tabla de valores

☐ **ordered pair**
par ordenado

☐ **axis**
eje

☐ **coordinates**
coordenadas

Model: *The number of tickets is the independent variable, and the independent variable always comes first in an ordered pair, so the point is (3, 135), not (135, 3).* — A strong answer names tickets as independent and explains that independent always goes first, then checks the arithmetic ($3 \times 45 = 135$). A weak answer just says 'because that's the order' without connecting it to independent/dependent.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- ____ is the independent variable, while ____ is the dependent variable. As the years go by, the interest earned ____, because each year adds \$ ____ to the total.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is** ____.
Mi afirmación es ____.
- **My evidence is** ____, **so** ____.
Mi evidencia es ____, *entonces* ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** graph
2. **Notes 2:** table of values
3. **Notes 3:** ordered pair
4. **Notes 4:** axis
5. **Notes 5:** coordinates
6. **Notes 6:** Rate of change
7. **Notes 7:** Origin
8. **Watch for this mistake:** *Reversing the coordinates of a point — writing (cost, tickets) instead of (tickets, cost). The independent variable always comes first in the ordered pair.*
9. **Try It 1:** C. 6,000 feet — *The remaining distance is $18,000 - 12,000 = 6,000$ feet — which matches the last 5 minutes at 1,200 feet per minute.*
10. **Try It 2:** C. The Sandia Peak line (1,200 ft/min) rises more steeply than the 900 ft/min line — *In one minute, the Sandia tram climbs 1,200 feet on the graph while the other climbs only 900 — a faster rate makes a steeper line.*